

ISIT912 Big Data Management
Assignment 2
Published on 25 August 2025

Scope

This assignment includes the tasks related to intuitive design of data cubes, conceptual modelling of data warehouse, implementation of "snowflake schema" as a collection of external tables in HQL and implementation of ONF Hive tables.

This assignment is due on **Saturday, 27 September 2025, 7:00pm** (sharp).

This assignment is worth **20%** of the total evaluation in the subject.

The assignment consists of 4 tasks and specification of each task starts from a new page.

Only electronic submission through Moodle at:
<https://moodle.uowplatform.edu.au/login/index.php>
will be accepted. A submission procedure is explained at the end of Assignment 1 specification.

A policy regarding late submissions is included in the subject outline.

Only one submission of Assignment 2 is allowed and only one submission per student is accepted.

A submission marked by Moodle as "late" is always treated as a late submission no matter how many seconds it is late.

A submission that contains an incorrect file attached is treated as a correct submission with all consequences coming from the evaluation of the file attached.

All files left on Moodle in a state "Draft (not submitted) " will not be evaluated.

A submission of compressed files (zipped, gzipped, rared, tared, 7-zipped, lhzed, ... etc) is not allowed. The compressed files will not be evaluated.

An implementation that does not compile well due to one or more syntactical and/or run time errors scores no marks.

Using any sort of Generative Artificial Intelligence (GenAI) for this assignment is NOT allowed !

It is expected that all tasks included within **Assignment 2** will be solved **individually without any cooperation** with the other students. If you have any doubts, questions, etc. please consult your lecturer or tutor during lab classes or office hours. Plagiarism will result in a **FAIL** grade being recorded for the assessment task.

Task 1 (2 marks)**Intuitive design of data cubes****(1) 0.3 mark**

Consider the following description of a sample database domain.

A transportation company collects information about the trips performed by its drivers. The company saves the collected data in a multidimensional data cube.

The company would like to use the collected data to compute an average fuel consumption per day, per month and per year, an average length of trips measured in kilometers per month and per year, and an average length of trips measured in hours.

Use a short explanation of a database domain given above to find a multi-dimensional data cube that should be implemented. In your specification of a data cube, list the facts, the measures, the names of dimensions and the hierarchies.

(2) 0.7 mark

Consider the following description of a sample database domain.

A transportation company collects information about the trips performed by its drivers. The company saves the collected data in a multidimensional data cube.

The company would like to use the collected data to compute an average fuel consumption per day, per month and per year, an average length of trips measured in kilometers per month and per year, and an average length of trips measured in hours.

After the successful implementation of the first stage, the company decided to extend the system. In the second stage, the company would like to compute the total number of trips per driver, per day, per month and per year and the total number of driving days per driver and year. Assume that a driver is described by his/her driving license number, first and last name.

Use a short explanation of a database domain given above to find a multi-dimensional data cube that should be implemented. In your specification of a data cube, list the facts, the measures, the names of dimensions and the hierarchies.

(3) 1.0 mark

Consider the following description of a sample database domain.

A transportation company collects information about the trips performed by its drivers. The company saves the collected data in a multidimensional data cube.

The company would like to use the collected data to compute an average fuel consumption per day, per month and per year, an average length of trips measured in kilometers per month and per year, and an average length of trips measured in hours.

After the successful implementation of the first stage, the company decided to extend the system. In the second stage, the company would like to compute the total number of trips per driver, per day, per month and per year and the total number of driving days per driver and year. Assume that a driver is described by his/her driving license number, first and last name.

After the successful implementation of the second stage, the company decided to extend the system again. In the third stage, the company would like to compute the total trips per different types of vehicles. The company owns cars, buses and trucks. A vehicle is described by registration number, make, model and fuel consumption. Additionally, the buses are described by the total seats available, and the trucks are described by the total capacity.

Use a short explanation of a database domain given above to find a multi-dimensional data cube that should be implemented. In your specification of a data cube, list the facts, the measures, the names of dimensions and the hierarchies.

Deliverables

A file `solution1.pdf` that contains

- (1) a specification of data cube including a list of facts, measures, dimensions, and hierarchies obtained as a solution of problem (1),
 - (2) a specification of data cube including a list of facts, measures, dimensions, and hierarchies obtained as a solution of problem (2),
 - (3) a specification of data cube including a list of facts, measures, dimensions, and hierarchies obtained as a solution of problem (3).
-

Task 2 (4 marks)

Conceptual modelling of a data warehouse

A large hospital would like to create a data warehouse to store information about the medical procedures performed on the patients and later to analyse the collected data. The planned data warehouse is designed to contain a large volume of historical data spanning many years.

A data warehouse supposed to contain information about the medical procedures, patients and medical staff involved in the medical procedures.

The hospital occupies several buildings and it consists of several wards located in the buildings. A building may host many wards and a ward must be entirely located in a single building. A building is described by building number, total number of rooms and total number of floors. A ward is described by a unique name.

Medical procedures are performed at the hospital. A medical procedure is described by its name, start date and time, end date and time and severity. A medical procedure is performed by a medical team that consists of doctors and nurses. A medical team is described by a unique name and rank. Each doctor and nurse belong to one medical team.

A medical procedure is performed at an operation theatre. A room number, building level, and building number describe an operation theatre. Each operation theatre is located at a ward. A ward may host many operation theatres. The hospital has a number of different types of operation theatres.

Patients are described by a medical insurance number, first name, last name, date of birth and address. An address consists of city name, street name building number and optional flat number. A medical insurance number identifies each patient.

Medical staff members are described by a unique employee number, first name, last name, and date of birth. A medical staff member belongs to one of the wards within a hospital. A medical staff member is either a nurse or a doctor. Nurses are additionally described by qualifications and doctors are described by a specialization.

Patients must pass through a medical examination just before a medical procedure. A medical examination is described by a date and time when performed and the measurement of the parameters like weight, blood pressure and body temperature.

A data warehouse must be designed such it would be possible to easily implement the following classes of applications.

- (1) *Find the total number of medical procedures and the average number of medical procedures performed per day, per month and per year.*

- (2) *Find the total length and the average length of medical procedures performed per day, per month and per year.*
- (3) *Find the total number of medical procedures and the average number of medical procedures performed per medical team, per doctor and per nurse.*
- (4) *Find the total number of patients per city they come from.*
- (5) *Find the average weight, blood pressure, body temperature per patient before medical procedure.*
- (6) *Find the total time and the average time of medical procedures per operation theatre.*
- (7) *Find the total number and the average number of medical procedures per ward.*
- (8) *Find the total number of medical procedures per type of operation theatre.*
- (9) *Find the total number of medical procedures per patient, per city patients come from.*
- (10) *Find the total number of medical examinations per day, per month per year.*

Your task is to create a conceptual schema for a data warehouse needed by the hospital.

To create a conceptual schema, follow the steps listed below.

- Step 1 Find a fact entity,
- Step 2 Find the measures describing a fact entity.
- Step 2 Find the dimensions.
- Step 3 Find the hierarchies over the dimensions.
- Step 4 Find the descriptions (attributes) of all entity types.
- Step 5 Draw a conceptual schema.

To draw a conceptual schema, use a graphical notation explained to you in a presentation Conceptual Data Warehouse Design. Use diagram drawing tool UMLet 15.1 and apply the graphical widgets available in ISIT312palette. The choice of the graphical widgets is available in the right upper corner of the main menu of UMLet.

UMLet 15.1 software is can be downloaded from a section Resources available at Moodle site for the subject.

Save a drawing of your conceptual schema in a file `solution2.bmp`.

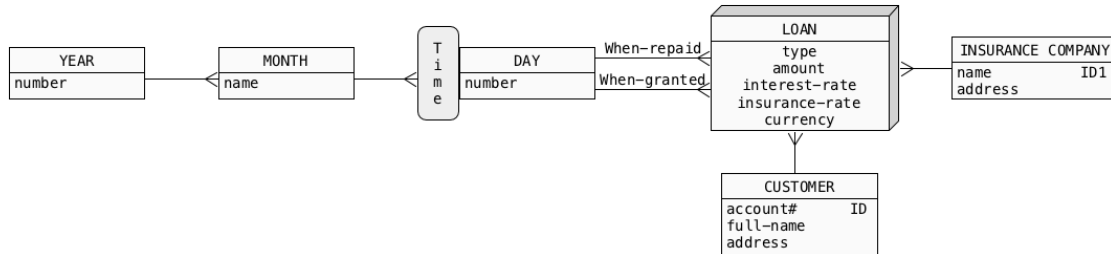
Deliverables

A file `solution2.bmp` with a drawing of a conceptual schema of a sample data warehouse domain.

Task 3 (6 marks)

Implementation of a data warehouse as a collection of internal tables in Hive

Consider the following conceptual schema of three-dimensional data cube.



(1) 2 marks

Perform a step of logical design and draw a "snowflake schema" obtained from the transformation of a conceptual schema given above.

When creating a "snowflake schema" apply the surrogate keys to implement the relationships. Clearly identify primary, candidate and foreign keys in the relational schemas. Reduce the number of dimensions to two in a "snowflake schema" through the implementation of time dimension as a single attribute.

To draw a "snowflake schema", use a graphical notation explained to you in a presentation 11 Logical Data Warehouse Design. To draw a "snowflake schema" use diagram drawing tool UMLet 15.1 and apply the graphical widgets available in LogicalDesign palette.

UMLet 15.1 software can be downloaded from a section Resources available at Moodle site for the subject.

Save a drawing of your "snowflake schema" in a file `solution3.bmp`.

(2) 4 marks

Implement a fact table and the dimension tables as the internal tables in Hive.

Create a script file `solution3.hql` with CREATE TABLE statements, LOAD statements to load sample data into the internal tables and SELECT statements that list the contents of each table. Do not load more than 5 rows per table. All input data are up to you.

When ready connect to Hive through beeline, process your script and save a report in a file `solution3.txt`.

Processing of your script must return NO ERRORS! **A solution with errors is worth no marks!**

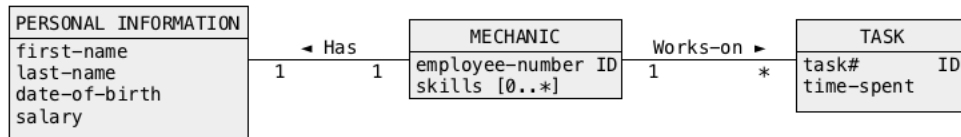
Deliverables

A file `solution3.bmp` with a drawing of "snowflake schema" of a data warehouse and a file `solution3.txt` with a report from processing of HQL script `solution3.hql`.

Task 4 (8 marks)

Implementation of 0NF table in Hive

Conceptual schema given below describes a database domain where mechanics have personal records and participates in the repairs of vehicles.



- (1) Implement HQL script `solution4.hql` that creates an internal 0NF relational table to store information about mechanics, their personal records and the tasks they work on. Use `array`, `map` and `struct` constructors for implementation of the table.
- (2) Include into the script `INSERT` statements that load sample data into the table. Insert at least 5 rows into the relational table created in the previous step. Two mechanics must work on some tasks and must have some skills. One mechanic must work on few tasks and must have no skills. One mechanic must have some skills and must not work on any task. One employee must have no skills and must not work on any task. All mechanics must have a nonempty employment record.
- (3) Include into the script `SELECT` statements that lists the contents of the table.

When ready, use a command interface `beeline` to process a script `solution4.hql` and to save a report from processing in a file `solution4.txt`.

If the processing of the file returns the errors then you must eliminate the errors! Processing of your script must return **NO ERRORS! A solution with errors is worth no marks!**

Deliverables

A file `solution4.txt` with a report from processing of HQL script `solution4.hql`. The report **MUST NOT include any errors**, and the report **must list all HQL statements processed**.

Task 5 (10 marks)

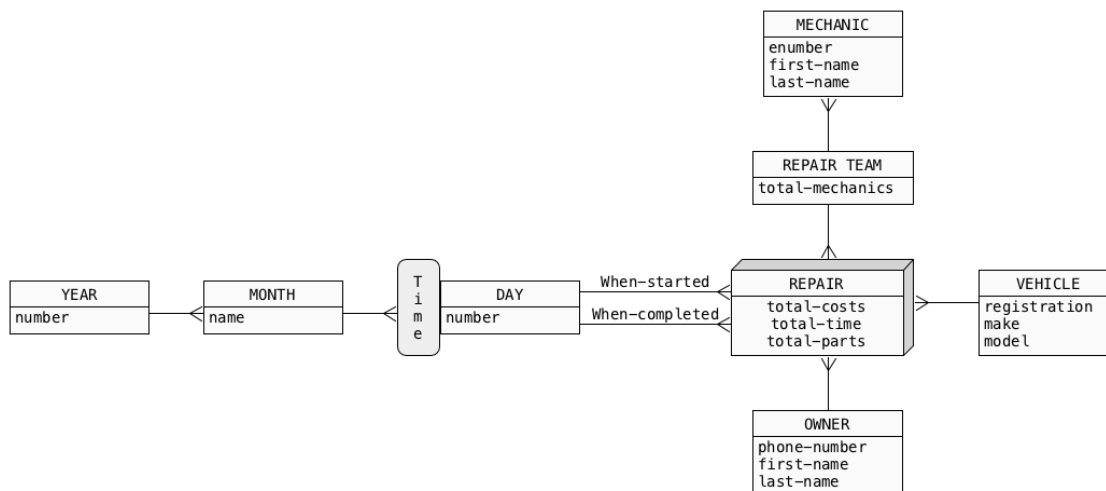
Logical design and loading data into a data warehouse database

A vehicle repair facility uses an operational database to keep information about the vehicles being repaired at the facility, mechanics involved in the repairs and owners of the vehicles. A database consists of the following tables.

```
MECHANIC(enumber, first-name, last-name, registration, start-date,  
         expected-end-date, status, repair-team)  
PART-USED(registration, part, cost)  
VEHICLE(registration, make, model, owner-phone)  
OWNER(owner-phone, first-name, last -name)
```

Whenever a vehicle is brought by an owner for the repairs and a repair team is assigned to a repair task, the new rows are inserted into MECHANIC table. A repair team consists of one or more mechanics. The new rows are also inserted into VEHICLE and OWNER tables. If a repair team consists of more than one mechanic then more than one row is inserted into MECHANIC table. Whenever a part is used a new row is inserted into PART-USED table. An initial value of attribute status is 'incomplete'. When all mechanics in a team working on a repair of the same vehicle complete their tasks the value of attribute status is changed to 'completed'. When a bill for a repair is paid all rows related to the repair are removed from all tables. It means that the database keeps only information about the ongoing and present repairs.

The vehicle repair facility would like to create a data warehouse to keep information about their past activities. A conceptual schema of the planned data warehouse database is given below.



(1) 2 marks

Perform a step of logical design and draw a "snowflake schema" obtained from the transformation of conceptual schema given above.

When creating a "snowflake schema" apply the surrogate keys to implement the relationships. Clearly identify primary, candidate and foreign keys in the relational schemas. A method used for implementation of time dimension is up to you.

To draw a "snowflake schema", use a graphical notation explained to you in a presentation 11 Logical Data Warehouse Design. To draw a "snowflake schema" use diagram drawing tool UMLet 15.1 and apply the graphical widgets available in LogicalDesign palette.

UMLet 15.1 software can be downloaded from a section Resources available at the Moodle site for the subject.

Save a drawing of your "snowflake schema" in a file `solution5.bmp`.

(2) 8 marks

When all mechanics working on the repairs of the same vehicle complete their tasks and a bill is paid all information related to the repairs of the same vehicle must be transferred to a data warehouse database before it is removed from all tables of an operational database. The objective of the second step is to provide detailed explanations on how data must be transferred from an operational database to a data warehouse database created in the previous step.

Consider a situation when all repairs on a given vehicle are completed and value of attribute `status` is 'complete' in all rows representing the mechanics that belong to a team involved in the repair. At this point we would like to transfer all information related to such repairs to a data warehouse database. In your solution provide all details of such transfers. You do not need to write any software, pseudo code, draw any BPM diagrams, etc. The comprehensive and precise explanations in English will do. Try to organize your answer according to different types of information, like information about a vehicle, mechanics, parts used, etc, being transferred to a data warehouse database.

Save your explanations in a file `solution5.pdf`.

Deliverables

A file `solution5.bmp` that contains a "snowflake schema" obtained from application of a logical design to a conceptual schema of data warehouse database.

A file `solution5.pdf` that contains the detailed and precise explanations on how data from an operational database must be transferred to a data warehouse database.

Submission of Assignment 2

Note that you have only one submission. So, make sure that you submit the correct files with the correct contents. No other submission is possible !

Submit the files **solution1.pdf**, **solution2.bmp**, **solution3.bmp**, **solution3.txt**, **solution4.txt**, **solution5.bmp** and **solution5.pdf** through Moodle in the following way:

- (1) Access Moodle at **<http://moodle.uowplatform.edu.au/>**
- (2) To login use a **Login** link located in the right upper corner the Web page or in the middle of the bottom of the Web page
- (3) When logged select a site **ISIT912/312 (S225) Big Data Management**
- (4) Scroll down to a section **Assessment items (Assignments)**
- (5) Click at **In this place you can submit the outcomes of your work on the tasks included in Assignment 2 for ISIT912 students** link.
- (6) Click at a button **Add Submission**
- (7) Move a file **solution1.pdf** into an area **File submissions**. You can also use a link **Add...**
- (8) Repeat a step (7) for the files **solution2.bmp**, **solution3.bmp**, **solution3.txt**, **solution4.txt**, **solution5.bmp** and **solution5.pdf**.
- (9) Click at a button **Submit assignment**.
- (10) Click at the checkbox with a text attached: **By checking this box, I confirm that this submission is my own work, I accept responsibility for any copyright infringement that may occur as a result of this submission, and I acknowledge that this submission may be forwarded to a text-matching service.**
- (11) Click at a button **Continue**
- (12) Check if **Submission status** is **Submitted for grading**.

End of specification